Resampling-Based Model Selection for PE SDFs

How to empirically measure the risk-adjusted performance of non-traded cash flows

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1 Introduction

Which stochastic discount factor (SDF) model gives the most credible risk adjustment for private-equity cash flows? This paper argues that the answer depends critically on a question that existing work treats as settled: *how should the pricing moments be constructed?*

The core problem: moment construction is consequential. Private-equity samples are small, lumpy, and organized by vintage year. An SDF model can look good or bad depending on how funds are grouped into pricing moments. Existing approaches commit to one grouping—typically single funds (e.g., Ang et al. [2018]) or vintage-year portfolios (e.g., Driessen et al. [2012])—and treat it as fixed. But the choice of grouping is itself consequential for estimation, and there is no a priori reason to prefer one admissible grouping over another.

We propose to make the grouping itself part of the empirical design. We repeatedly draw random within-vintage fund partitions, estimate the SDF on partition-induced moments, and evaluate pricing errors on held-out moments. The preferred SDF model is the one with the lowest average holdout pricing loss across many admissible partitions. We call this *resampling-based model selection*: moment construction becomes a testable design choice rather than a fixed input.

Two additional difficulties arise within this framework and must be handled by modular design choices:

Unseasoned vintages create a dilemma. Recent vintage years carry valuable pricing information—they reflect current market conditions and expand the factor-path coverage of the sample—but including them naively destabilizes (or biases) estimation. The standard approach treats the latest reported NAV as a terminal cash flow and discounts it back to the fund inception date. This creates two problems. First, it concentrates a large share of the fund's discounted value at a single calendar date—the dataset endpoint—so that the estimator's fit becomes disproportionately sensitive to the macro state at that date rather than to the fund's lifetime factor exposure. Second, the terminal NAV is an appraisal, not a realized payoff; its co-movement with public indices may differ systematically from that of actual divestitures. Researchers therefore face a dilemma: exclude recent vintages and waste data, or include them and accept fragile estimates driven by one macro-state realization. Rather than commit to one approach, our framework accommodates and empirically compares three competing strategies for unseasoned vintages (Section 6).

Exploding-alpha degeneracy in finite samples. NPV-style SDF estimation suffers from a finite-sample pathology first documented by Driessen et al. [2012]. In the standard multiplicative SDF parameterization, cash flows are discounted by a cumulative factor that includes $(1 + \alpha)^{-T}$. When α is large and positive, this factor drives all discount factors toward zero, making every fund's pricing error approximately zero regardless of the factor loadings β. The result is a ridge of spurious near-minima along the alpha axis of the objective function. The problem is most severe when fund horizons are long and early cash flows are small relative to later ones, which is precisely the cash-flow shape of a typical PE fund. Validation folds are useful for model comparison, but they do not by themselves solve this pathology: if the candidate cumulative SDF collapses all long-horizon discounted cash flows, a held-out NPV moment can collapse as well. We therefore treat exploding alpha as a separate identification and stability issue, handled through alpha design, economically admissible parameter restrictions, and diagnostics for boundary solutions (Section 7).

Summary of the approach. The method has three interrelated goals:

- **Model selection:** rank SDF models by out-of-sample pricing performance across many resampled moment constructions.
- **Design comparison:** compare whether within-vintage fund resampling or traditional vintage-year portfolio construction yields more stable and predictive pricing moments.
- **Efficient use of the vintage panel:** include all available vintages—seasoned and unseasoned—through flexible, smart resampling designs tailored to different approaches for handling unseasoned data.

The paper is closest in spirit to Korteweg and Nagel [2016] because the estimator works with aggregated pricing moments, and to Driessen et al. [2012] because the moments price private cash flows directly. The new contribution is a general resampling-based model selection framework that (i) treats moment construction as a testable design choice, (ii) nests competing approaches to unseasoned vintages as special cases and ranks them empirically, and (iii) can accommodate time-varying factor specifications through vintage-stratified estimation (Section 9). Brown et al. [2023] also use NAVs to estimate the risk and return of PE funds, and Ghysels et al. [2025] construct high-frequency PE returns from NAV-based state-space models and use them for factor analysis and spanning tests.

2 Guiding Example

Suppose a candidate SDF has three parameters: an alpha term and two factor loadings. Then we need at least three pricing moments to estimate it. A natural resampling design is therefore:

- build three estimation folds,
- build one validation fold,
- estimate the SDF on the three estimation moments,
- score the fitted SDF on the validation moment (calculate out-of-sample pricing error).

In stylized notation, the example can be summarized with one simple three-parameter SDF and one resampled moment per fold. Let the parameter vector be

$$\theta = (\alpha, \beta_1, \beta_2)',$$

and let the one-period discount factor be

$$M_{t-1,t}(\theta) = \frac{1}{1 + \alpha + \beta_1 f_{1,t} + \beta_2 f_{2,t}}, \qquad \Psi_{0,t}(\theta) = \prod_{s=1}^{t} M_{s-1,s}(\theta),$$

where $f_{1,t}$ and $f_{2,t}$ are two benchmark factor realizations. For transparency the running example uses the additive period-by-period intercept form of α; this is the parameterization we later label DLP12-style and flag as the most exposed to exploding alpha (Section 8). The recommended baseline instead uses a cumulative, NPV-style alpha, and the resampling machinery is identical for either choice. For sampled split n, let $C_{n,v,r}$ be the funds from vintage v assigned to fold $r \in \{1, 2, 3, 4\}$, with $r = 1, 2, 3$ used for estimation and $r = 4$ used for validation. Write $CF_{t,i} = D_{t,i} - X_{t,i}$ for net cash flow, let K_i denote fund i's committed capital, and partition active vintages into seasoned and unseasoned sets, $\mathcal{V} = \mathcal{V}^S \cup \mathcal{V}^U$. For the unseasoned block, let

$R^H_{t,i}$ denote the gross quarterly Horizon-IRR return computed from beginning NAV, ending NAV, and dated capital calls and distributions within quarter t. The vintage-level pricing restriction is then

$$\epsilon_{n,v,r}(\theta) = \begin{cases} \epsilon^S_{n,v,r}(\theta) = \dfrac{1}{|C_{n,v,r}|} \sum_{i \in C_{n,v,r}} \dfrac{1}{K_i} \sum_{t=0}^{T_i} \Psi_{0,t}(\theta)\, CF_{t,i}, & v \in \mathcal{V}^S, \\ \epsilon^U_{n,v,r}(\theta) = \dfrac{1}{|C_{n,v,r}|} \sum_{i \in C_{n,v,r}} \Lambda_i \dfrac{1}{T_i} \sum_{t=1}^{T_i} e^H_{t,i}(\theta), & v \in \mathcal{V}^U, \end{cases}$$

where the unseasoned one-period Euler error is

$$e^H_{t,i}(\theta) = M_{t-1,t}(\theta) R^H_{t,i} - 1.$$

Here $e^H_{t,i}(\theta)$ is already a relative quarterly pricing error, while Λ_i is a deterministic duration-matching normalizer based on fund age and strategy (Section 6.3) that maps the fund-average relative Euler error into an NPV-like quantity per committed dollar. The resampled fold moment is the cross-vintage average

$$m_{n,r}(\theta) = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \epsilon_{n,v,r}(\theta).$$

Because θ has three components, we write three stylized scalar fold conditions for exposition,[1]

$$m_{n,1}(\theta) = 0, \qquad m_{n,2}(\theta) = 0, \qquad m_{n,3}(\theta) = 0,$$

or equivalently

$$\hat{\theta}_n = \arg\min_{\theta} \sum_{r=1}^{3} m_{n,r}(\theta)^2.$$

The held-out fourth fold then gives the validation score

$$S_n(\hat{\theta}_n) = m_{n,4}(\hat{\theta}_n)^2.$$

Our resampling-based approach repeats this exercise many times: for each draw $n = 1, \ldots, N$, we redraw the partition and thereby obtain a new quartet of fold moments $\{m_{n,1}, m_{n,2}, m_{n,3}, m_{n,4}\}$.

Figures 1 and 2 show the two baseline ways to build these folds. In both figures, each dot is one fund. Vintages 2005–2010 lie above the approach boundary and use cash-flow discounting, while vintages 2011–2012 lie below it and use the normalized Euler approach.

[1] This scalar-per-fold device is only illustrative. As Section 5.2 shows, exchangeable folds share a single population moment, so the parameters are in fact identified by the cross-section of vintages *within* each fold, not by the three folds; the fold count fixes the estimation/validation split rather than the number of identifying moments.

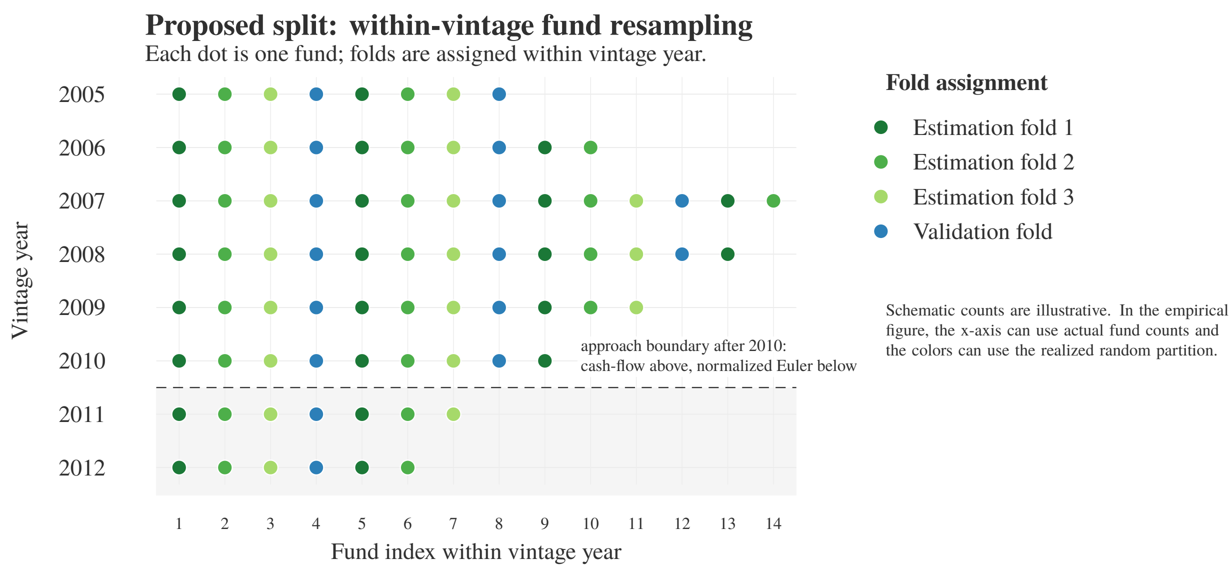

Figure 1: Proposed within-vintage fund resampling scheme. Each active vintage is split into three estimation cells and one validation cell. Fold 1 contains the dark-green funds from every vintage, fold 2 contains the medium-green funds from every vintage, fold 3 contains the light-green funds from every vintage, and the validation fold contains the blue funds from every vintage.

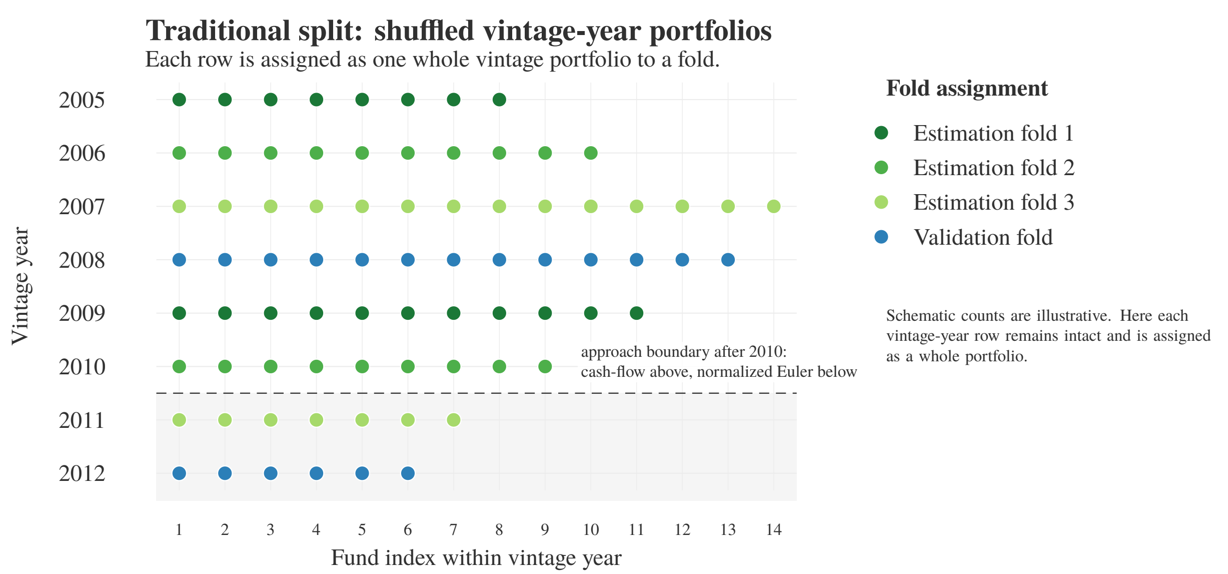

Figure 2: Traditional shuffled vintage-year portfolio scheme. Funds are first aggregated into whole vintage-year portfolios, and whole vintages are then assigned to folds. Here, 2005 and 2009 enter estimation fold 1, 2006 and 2010 enter estimation fold 2, 2007 and 2011 enter estimation fold 3, and 2008 and 2012 form the validation fold.

3 Framework and Notation

Let $v = 1, \ldots, V$ index active vintage years, and let $\mathcal{I}_v$ be the set of funds in vintage v. Let M denote a candidate SDF model with parameter vector $\theta \in \Theta_M \subset \mathbb{R}^{p(M)}$.

We use P estimation folds and h holdout folds. For the guiding three-parameter example, $P = 3$ and $h = 1$. When comparing models with different numbers of parameters, a clean implementation is to use a common number of estimation folds $P = p_{\max}$, where $p_{\max} = \max_M p(M)$. As Section 5.2 explains, P governs the holdout architecture and cross-model comparability, not the identification of θ, which comes from the within-fold vintage cross-section.

3.1 Within-vintage resampling

In the proposed design, every fold contains funds from every active vintage. This keeps the vintage-year support fixed across estimation and validation moments. The folds differ because they contain different funds, not because they represent different macro or vintage states. This is the cleanest design for asking whether an SDF prices PE cash flows robustly across fund-level resamples.

For sampled split $n = 1, \ldots, N$, split the funds within each vintage into $P + h$ cells:

$$\mathcal{P}_{n,v} = \{C_{n,v,1}, \ldots, C_{n,v,P+h}\}, \qquad C_{n,v,r} \subset \mathcal{I}_v.$$

The cells are disjoint and cover the funds in vintage v. The rth cross-vintage fold is

$$F_{n,r} = \{C_{n,1,r}, C_{n,2,r}, \ldots, C_{n,V,r}\}.$$

Thus fold r contains one fund sub-sample from every active vintage.

At the single-fund level, the number of admissible assignments is usually too large to enumerate. We therefore draw Monte Carlo samples from a partition rule Π:

$$\mathcal{P}_n \sim \Pi(\cdot \mid P, h, \mathcal{I}_1, \ldots, \mathcal{I}_V).$$

For now, Π should be simple and transparent: random assignment within each vintage, subject to no empty cells and reasonable balance in fund counts or economic size.

3.2 Vintage-year portfolio benchmark

In the traditional design, the primitive object is a vintage-year portfolio. A fold contains whole vintages, not fund sub-samples from all vintages. This is easy to explain and resembles standard PE portfolio construction. But the validation moment now differs from the estimation moments in vintage composition. A failure on the validation fold may therefore reflect poor SDF pricing, or it may reflect extrapolation across vintage states.

The notation is the same at the fold level, but the economic object differs: $F_{n,r}$ now contains whole vintage-year portfolios rather than within-vintage fund cells.

3.3 Design comparison

The first version of the paper should focus on this contrast. If within-vintage resampling delivers more stable estimates and lower holdout pricing errors, the evidence favors building all-vintage subportfolios before estimating the SDF. If shuffled vintage-year portfolios perform similarly, the simpler vintage-year construction may be sufficient.

4 Pricing Moments

For a fold $F_{n,r}$, let $\epsilon_{n,v,r}(\theta; M, \eta)$ be the pricing error from the vintage component of that fold. The design choice η collects choices such as the alpha definition, loss function, and cash-flow weighting.

The aggregated fold moment is

$$m_{n,r}(\theta; M, \eta) = \sum_{v \in A_{n,r}} \omega_{n,v,r}\, \epsilon_{n,v,r}(\theta; M, \eta),$$

where $A_{n,r}$ is the set of vintages represented in fold r, and $\omega_{n,v,r}$ are vintage or value weights. Under within-vintage resampling, $A_{n,r}$ is the full active vintage set for every fold. Under shuffled vintage-year portfolios, $A_{n,r}$ is only the set of whole vintages assigned to fold r.

This distinction is the core empirical design choice. The proposed method creates comparable fold moments that all span the same vintage domain. The benchmark creates moments from different vintage domains.

Scalar versus vector fold losses. The baseline collapses the vintage components into one scalar fold moment $m_{n,r}$. This is simple and keeps the number of moments close to the number of parameters, but it can hide sign cancellation: overpricing one vintage can offset underpricing another. A natural refinement is to keep the vintage or strategy-vintage components as a vector,

$$\boldsymbol{\epsilon}_{n,r}(\theta) = \left(\epsilon_{n,v,r}(\theta) : v \in A_{n,r}\right), \qquad L^{\text{vec}}_{n,r}(\theta) = \boldsymbol{\epsilon}_{n,r}(\theta)' W_{n,r} \boldsymbol{\epsilon}_{n,r}(\theta),$$

where $W_{n,r}$ is diagonal, shrinkage, or block-cluster weighted. This does create more moment conditions in the GMM sense if used for estimation, so it may require regularization or a model with fewer free parameters. As a first pass, the scalar moment can remain the baseline and the vector loss can be used as a diagnostic score: does the model price each vintage or strategy-vintage cell, not just their average?

4.1 Fund weights and vintage weights

The fund-level scaling and vintage weight are design choices that determine the economic content of the fold moment. For seasoned funds, the baseline contribution is

$$g_i^S(\theta; M, \eta) = \frac{1}{K_i} \sum_{t=0}^{T_i} \Psi_{0,t}(\theta; M, \eta)\, CF_{t,i},$$

so each seasoned fund contributes a *relative* NPV error per dollar committed. For unseasoned funds, the analogous baseline contribution is the normalized Euler error $g_i^U(\theta; M, \eta)$ defined in Section 6.3: a fund-average one-period Horizon-IRR Euler error rescaled by a deterministic duration factor. At the cell level, we average over the funds in the cell. At the vintage level, the baseline sets $\omega_{n,v,r} = 1/|A_{n,r}|$ (equal vintage weights), so that each vintage contributes equally to the fold moment regardless of the number of funds or aggregate capital it contains.

Under this hybrid three-layer normalization (fund $\to$ cell $\to$ vintage), the fold moment becomes

$$m_{n,r}(\theta; M, \eta) = \frac{1}{|A_{n,r}|} \left[\sum_{v \in \mathcal{V}^S \cap A_{n,r}} \frac{1}{|C_{n,v,r}|} \sum_{i \in C_{n,v,r}} g_i^S(\theta; M, \eta) \;+\; \sum_{v \in \mathcal{V}^U \cap A_{n,r}} \frac{1}{|C_{n,v,r}|} \sum_{i \in C_{n,v,r}} g_i^U(\theta; M, \eta) \right].$$

This is an average, across vintages, of the average relative pricing error per fund. For seasoned vintages the object is a relative NPV error; for unseasoned vintages it is a duration-matched relative Euler error expressed in comparable per-committed-dollar units. It is therefore comparable across: (i) vintages with different numbers of funds, (ii) vintages with different aggregate capital, (iii) seasoned vintages with cash-flow errors and unseasoned vintages with normalized Euler errors (Section 6.3), and (iv) estimation folds and validation folds.

4.2 Choice of pricing moment

The pricing error in the loss L is a design choice, not a primitive. Two canonical families appear in the private-capital SDF literature: the *dollar NPV* moment of Driessen et al. [2012] and the *PME-ratio* moment of Kaplan and Schoar [2005], Driessen et al. [2009], and Ang et al. [2018]. We treat the NPV moment as the baseline and the PME-ratio as a robustness check.

NPV moment (baseline). For seasoned funds, aggregate discounted cash flows into one cumulative pricing error per dollar committed. For unseasoned funds, use the normalized Euler fund moment $g_i^U(\theta; M, \eta)$ from Section 6.3. The empirical baseline is therefore hybrid: cumulative NPV for $\mathcal{V}^S$, normalized Euler for $\mathcal{V}^U$. The NPV moment is linear in cash flows, additive across funds, and compatible with the normalized Euler NAV restriction for unseasoned vintages. Both blocks are expressed as per-committed-dollar pricing errors, and neither requires ratio denominators that can approach zero.

PME-ratio moment (robustness check). Define the risk-adjusted PME of fund i as the ratio of discounted distributions to discounted contributions. This is scale-invariant and has natural practitioner interpretability, but it is not the same statistical object as the NPV moment. If $D_i^*(\theta)$ and $X_i^*(\theta)$ denote discounted distributions and contributions, the NPV restriction is $E[D_i^*(\theta_0) - X_i^*(\theta_0)] = 0$. It does not generally imply $E[D_i^*(\theta_0)/X_i^*(\theta_0)] = 1$, because the expectation of a ratio is not the ratio of expectations and depends on the joint distribution of numerator and denominator. The average-PME and pooled-PME moments therefore estimate different population objects and require separate interpretation. The PME ratio is also nonlinear in θ, numerically fragile when the denominator is small, and has no clean short-horizon analogue for unseasoned vintages—integrating $\mathcal{V}^U$ into a PME framework typically requires treating the latest NAV as a terminal distribution and discounting it back to the fund inception date, reintroducing the long-horizon instability the normalized Euler block is designed to avoid. We therefore restrict the PME robustness check to seasoned vintages $\mathcal{V}^S$. Detailed pros and cons of each moment family are given in Appendix A.

Robustness diagnostic. Stability of the preferred model-design pair (M, η, Π) across the hybrid NPV/Euler baseline, the raw dollar NPV, and the pooled-PME moment (restricted to $\mathcal{V}^S$) is the appropriate robustness diagnostic: agreement suggests genuine model fit, while disagreement would flag moment-induced identification.

5 Estimation and Holdout Scoring

For each sampled split n, let $\mathcal{E} = \{1, \ldots, P\}$ be the estimation-fold indices and let $\mathcal{H} = \{P+1, \ldots, P+h\}$ be the holdout-fold indices. In the baseline case $h = 1$, write $r_H = P + 1$ for the single holdout fold. The split-specific estimator solves

$$\hat{\theta}_n(M, \eta, \Pi) = \arg\min_{\theta \in \Theta_M} \frac{1}{P} \sum_{j \in \mathcal{E}} L\big(m_{n,j}(\theta; M, \eta)\big) .$$

The validation score is the pricing loss on the held-out fold(s):

$$S_n(M, \eta, \Pi) = \frac{1}{|\mathcal{H}|} \sum_{r \in \mathcal{H}} L\Big(m_{n,r}(\hat{\theta}_n(M, \eta, \Pi); M, \eta)\Big) .$$

Aggregate this score across many sampled assignments:

$$\text{Score}_N(M, \eta, \Pi) = \frac{1}{N} \sum_{n=1}^{N} S_n(M, \eta, \Pi).$$

The preferred model-design pair (M, η, Π) is the one with the lowest average holdout pricing loss. Parameter stability is still informative, but the primary ranking criterion should be predictive pricing performance: does the fitted SDF price cash flows that were not used in estimation?

5.1 Why validation is needed

If a model has p parameters and we estimate it on exactly p moments, the in-sample fit can look deceptively good. This is especially dangerous in small PE samples because exact identification can hide instability. The validation fold creates a more demanding model-selection question: after fitting the model on P moments, does it also price a held-out PE cash-flow portfolio? This validation exercise should not be interpreted as a cure for the exploding-alpha pathology; that issue requires the separate identification and stability checks in Section 7.

5.2 Identification: the folds are not the identifying moments

It is tempting to read the guiding example as "$p(M)$ parameters, therefore $P = p(M)$ estimation folds, therefore $p(M)$ moment conditions." This counting is misleading. Under within-vintage resampling the P estimation folds are *exchangeable subsamples of the same fund populations*, and exchangeable moments are redundant for identification in the precise GMM sense. We keep within-vintage resampling as the design, but it is important to be explicit about what does—and does not—identify the SDF parameters.

Fix a split n and recall the fold moment

$$m_{n,r}(\theta) = \frac{1}{|A_{n,r}|} \sum_{v \in A_{n,r}} \epsilon_{n,v,r}(\theta), \qquad \epsilon_{n,v,r}(\theta) = \frac{1}{|C_{n,v,r}|} \sum_{i \in C_{n,v,r}} g_i(\theta).$$

Within a vintage, the cells $C_{n,v,1}, \ldots, C_{n,v,P}$ are a random partition of the same funds, so each fund-cell average is an unbiased draw of the vintage population pricing error

$$\bar{g}_v(\theta) = E[g_i(\theta) \mid i \in \text{vintage } v], \qquad E\big[\epsilon_{n,v,r}(\theta)\big] = \bar{g}_v(\theta) \ \text{ for every } r.$$

Hence every estimation fold has the *same* population moment,

$$E\big[m_{n,r}(\theta)\big] = \frac{1}{|A_{n,r}|} \sum_{v \in A_{n,r}} \bar{g}_v(\theta) =: \bar{m}(\theta), \qquad r = 1, \ldots, P.$$

Stacking the P fold moments into $G(\theta) = \big(E[m_{n,1}(\theta)], \ldots, E[m_{n,P}(\theta)]\big)' = \bar{m}(\theta)\,\mathbf{1}_P$ yields a Jacobian that is an outer product,

$$\frac{\partial G(\theta_0)}{\partial \theta'} = \mathbf{1}_P\,\frac{\partial \bar{m}(\theta_0)}{\partial \theta'}, \qquad \operatorname{rank}\!\left(\frac{\partial G(\theta_0)}{\partial \theta'}\right) \le 1.$$

The local identification (rank) condition $\operatorname{rank}\big(\partial G(\theta_0)/\partial\theta'\big) = p(M)$ therefore fails for any $p(M) \ge 2$, no matter how many folds are used: the P folds restrict only the single scalar combination $\bar{m}(\theta) = 0$ and leave a $(p(M) - 1)$-dimensional set of observationally equivalent parameters. A pricing moment is *nonredundant* only if its population pricing error is not a linear combination of the others and its derivative with respect to θ adds independent rank; replicated exchangeable folds are duplicates on both counts.

The redundancy is structural, not a feature of any particular SDF. Folds are exchangeable, so they share one population moment and one Jacobian row. Vintages are *not* exchangeable: each vintage lives through a different factor path and cash-flow timing, so the sensitivities $\partial\bar{g}_v(\theta_0)/\partial\theta'$ differ across v. The identifying rank lives in the cross-vintage Jacobian

$$J(\theta_0) = \frac{\partial}{\partial\theta'}\big(\bar{g}_v(\theta_0)\big)_{v\in A} \;\in\; \mathbb{R}^{|A|\times p(M)}, \qquad \text{identification} \iff \operatorname{rank}\ J(\theta_0) = p(M),$$

i.e. at least $p(M)$ vintages (or strategy–vintage cells) with linearly independent pricing-error sensitivities. Collapsing each fold to a scalar replaces J by its column sum $\mathbf{1}'J$, a single row, discarding exactly the cross-vintage rank that identifies the factor loadings. This is the precise content of the scalar-versus-vector distinction in Section 4: the vintage (or strategy–vintage) component vector $\boldsymbol{\epsilon}_{n,r}(\theta)$ is not an optional refinement but the object that carries identification.

Two consequences for the empirical design follow.

Identification and resampling play orthogonal roles. The vintage cross-section *within* a fold supplies the rank condition; the partition *into* folds supplies the holdout split. We therefore retain within-vintage resampling, but the estimator at each split minimizes a loss over the within-fold vintage moment vector $\boldsymbol{\epsilon}_{n,r}(\theta)$ of Section 4, not over a single scalar per fold. The number of estimation folds is then governed by the desired holdout architecture, not by $p(M)$; the convention $P = p_{\max}$ should be read as a device for cross-model comparability, not as the source of identification.

A weak-identification diagnostic, not a counting rule. Because $P \ge p(M)$ moments can still be rank deficient, the relevant check is the rank or condition number of the estimated moment Jacobian at $\hat{\theta}_n$, not the moment count. In finite samples the scalar fold moments differ only through sampling noise in cell composition, so the between-fold Jacobian variation is $O(1/|C_{n,v,r}|)$ and vanishes as cells grow: the scalar-fold estimator is weakly identified in the sense of Stock and Wright [2000], and its apparent finite-sample precision is an artifact of that noise. We therefore report the Jacobian condition number alongside the cross-split dispersion of $\hat{\theta}_n$ and the boundary/non-convergence frequencies of Section 11 as identification diagnostics.

5.3 The resampling estimator as subagging

Averaging over sampled partitions is not merely a convenience: it is *subagging* (subsample aggregating) of the split-level estimator [Bühlmann and Yu, 2002]. Each draw n supplies a partition, the split-level GMM solution $\hat{\theta}_n$ is the base estimator, and the framework reports two bagged objects—a bagged parameter $\bar{\theta}_N = \frac{1}{N}\sum_n \hat{\theta}_n$ and the bagged score $\text{Score}_N = \frac{1}{N}\sum_n S_n$. Reading the method this way clarifies both what the averaging buys and what it cannot.

What it buys. Bagging smooths the discontinuous dependence of the base estimator on the partition and reduces its variance, with the largest gains for unstable estimators [Bühlmann and Yu, 2002]—exactly the regime of small PE samples with occasional boundary or near-degenerate solutions. The bagged score is a smoother, lower-variance model-selection criterion than any single split, and the paired-difference comparison of Section 11 operates on this bag.

What it cannot buy. Subagging stabilizes an estimator; it does not create identification. If the base estimator is built from exchangeable scalar fold moments, its split-level Jacobian is rank deficient (Section 5.2) and averaging many such solutions only produces a tight bag around a partition-induced pseudo-value—false precision. Subagging is therefore valuable *on top of* the vector-moment identification of Section 5.2, not as a substitute for it.

Closed-form partition moments for the linear baseline. For the baseline NPV/normalized-Euler moment, which is linear in the fund-level contributions $g_i(\theta)$ at fixed θ, the average over random within-vintage partitions has a closed form, so the Monte Carlo over Π is unnecessary for the fold *moment* itself. Let $\bar{g}_v(\theta) = |\mathcal{I}_v|^{-1} \sum_{i \in \mathcal{I}_v} g_i(\theta)$ be the within-vintage grand mean and $\sigma_v^2(\theta)$ the within-vintage cross-sectional variance of $g_i(\theta)$. Under a uniform random partition of $\mathcal{I}_v$ into equal cells, each cell mean is a simple random sample without replacement, so

$$\mathbb{E}_\Pi\left[\epsilon_{n,v,r}(\theta)\right] = \bar{g}_v(\theta), \qquad \mathrm{Var}_\Pi\left[\epsilon_{n,v,r}(\theta)\right] = \frac{\sigma_v^2(\theta)}{|C_{n,v,r}|}\left(1 - \frac{|C_{n,v,r}| - 1}{|\mathcal{I}_v| - 1}\right).$$

The cell mean is thus a one-sample U-statistic, and the partition-induced variance of the fold moment is governed by the within-vintage dispersion σ_v^2 and the finite-population correction. Three consequences follow. First, the baseline score can be reported against the exact grand-mean moment, with N controlling only the residual nonlinear pieces—the estimation map $\hat{\theta}_n$, a nonlinear loss L, or the PME ratio—rather than treated as a free tuning knob. Second, the variance formula gives an analytic target for how many draws N achieve a desired score precision. Third, because the noise scales with σ_v^2, balancing cells on the fund characteristics that drive g_i (size, strategy, duration)—antithetic or stratified partitions in place of uniform draws—reduces partition variance for free and tightens the paired model comparison.[2]

5.4 Two validation targets

The empirical implementation should distinguish two validation targets.

Within-vintage holdout validation. The baseline resampling design holds out funds within the same active vintage set. Estimation and validation folds therefore share the same vintage-year support but contain different funds. This target asks whether the fitted SDF prices fund-level cash-flow variation that was not used in estimation, conditional on the same vintage and macro-state support. Because estimation and validation folds share the same calendar and macro-state support, this is a *within-state* holdout: it does not test generalization across macro regimes, and in the terminology of dependent-data cross-validation it is not *hv*-block separated [Racine, 2000]. It is nonetheless the cleanest target for comparing SDF factor specifications and moment-construction rules within a given treatment of unseasoned vintages; the rolling-origin target below supplies the across-state counterpart.

[2]Concretely, within each vintage stratify funds into bins by the characteristics that drive g_i—committed size, strategy, and duration—and fill the $P + h$ cells by allocating bins as evenly as possible (a balanced or antithetic assignment) rather than by independent uniform draws. Each cell then has matched composition, so the between-cell variance of $\epsilon_{n,v,r}$ shrinks and the finite-population factor σ_v^2 is effectively replaced by the smaller within-stratum dispersion. This only changes the partition rule Π: the cells remain valid estimation/holdout splits, but the variance formula above must then use the stratified sampling variance, and any paired model comparison must hold Π fixed across the models being compared.

Rolling-origin / out-of-time validation. To compare unseasoned-vintage approaches, the cleanest target is a forward-looking rolling-origin design. Pick an historical information date τ. Using only information available at τ, classify then-seasoned and then-unseasoned vintages, estimate $\hat{\theta}_\tau^{(k)}$ under each approach $k \in \{A, B, C\}$, and record which vintages were treated as unseasoned at that date. Then move the evaluation date forward until those vintages have become seasoned, and score every approach on the same realized-cash-flow criterion:

$$S_\tau^{\text{roll}}(k) = L\left(\sum_{v \in \mathcal{V}_\tau^U \cap \mathcal{V}_{\tau+\ell}^S} \omega_{v,\tau+\ell}\, \epsilon_{v,\tau+\ell}^S \left(\hat{\theta}_\tau^{(k)}\right) \right).$$

This score asks which approach, when forced to use only the information available at τ, produces parameters that later price the realized cash flows of the same vintages once their NAV uncertainty has resolved. Repeating the exercise across rolling origins τ gives a paired, common-scale comparison of the unseasoned-vintage approaches.

The main empirical specifications should be:

- **Main comparison:** $P = p_{\max}$, $h = 1$, so all candidate models are compared on a common fold architecture.
- **Exact-identified diagnostic:** $P = p(M)$, $h = 0$, to study feasibility and parameter instability.
- **Model-specific validation:** $P = p(M)$, $h = 1$, to compare exact-identified estimation plus one holdout fold.
- **Scheme comparison:** run the same model list under within-vintage resampling and shuffled vintage-year portfolios.
- **Rolling-origin comparison:** estimate each unseasoned-vintage approach using historical information dates and score later on common realized-cash-flow moments.

6 Approaches to Unseasoned Vintages

Partition the active vintages into seasoned and unseasoned sets,

$$\mathcal{V} = \mathcal{V}^S \cup \mathcal{V}^U.$$

A vintage v is *unseasoned* ($v \in \mathcal{V}^U$) if its aggregate residual NAV is large relative to the value it has already realized; otherwise it is *seasoned* ($v \in \mathcal{V}^S$). A convenient scale-free criterion is the unrealized-value fraction RVPI/(RVPI + DPI) exceeding a threshold δ; scaling residual NAV by committed capital is an alternative, but that denominator can remain near or above one for strong funds long after they are effectively mature. In practice, $\delta \in [0.10, 0.20]$ is a sensible range. Thus "seasoned" does not require residual NAV to be exactly zero; it means the remaining NAV is small enough that the vintage is treated as effectively mature for the chosen empirical design. For seasoned vintages, the pricing restriction is the standard cumulative cash-flow error as defined in Section 4.

How to handle the small residual NAV of a seasoned vintage is itself a design choice in η. One can exclude residual NAV, include it as a final distribution, or apply a de minimis NAV adjustment. The important requirement is consistency: once a terminal-NAV convention is chosen for a specification, it must be applied identically in estimation folds, holdout folds, robustness checks, and common-score comparisons.

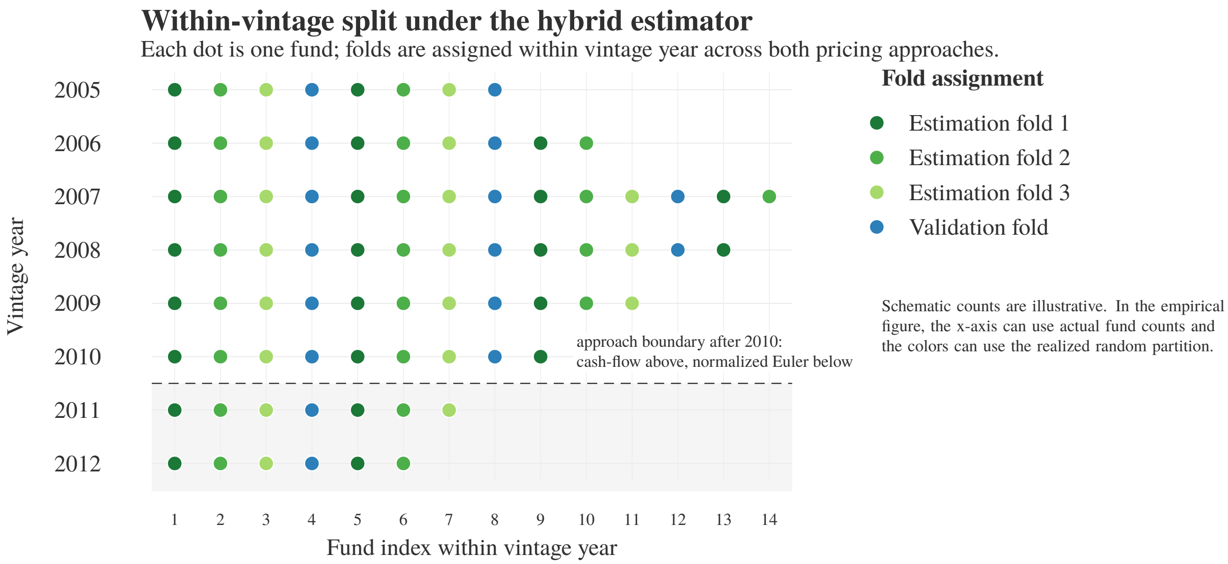

Figure 3: Within-vintage fold assignment under the hybrid cash-flow / normalized-Euler estimator. Unseasoned vintages 2011–2012 ($\mathcal{V}^U$, grey band) are included in estimation and validation like the seasoned vintages ($\mathcal{V}^S$), but contribute the normalized one-period Euler NAV error ϵ^U in place of the cumulative cash-flow error ϵ^S.

The central question is how to include unseasoned vintages—or whether to include them at all. Rather than commit to one approach a priori, our framework accommodates three competing strategies and lets the resampling-based holdout loss rank them empirically.

6.1 Approach A: NAV-adjusted cash-flow SDF

The most direct strategy adjusts the terminal NAV before applying the standard cumulative cash-flow SDF. For seasoned vintages, the pricing restriction is unchanged. For unseasoned vintages, the latest reported NAV is multiplied by an empirical discount factor d_i before being treated as a terminal distribution:

$$\widetilde{D}_{T_i,i} = D_{T_i,i} + d_i \cdot \mathrm{NAV}_{T_i,i},$$

where $d_i \in (0, 1]$ is the NAV discount. The fund then enters the standard CF-SDF pricing restriction using $\widetilde{D}_{T_i,i}$ as the terminal cash flow.

The discount d_i can be calibrated from empirical secondary-market transaction data—for example, from the Setter Capital secondary transaction dataset, which provides observed bid–ask spreads and transaction prices for LP interests as a function of fund age, strategy, and market conditions. This grounds the NAV adjustment in real transaction data rather than arbitrary assumptions.

Advantages. The approach is conceptually simple and uses the same cumulative SDF for all vintages, maintaining a unified pricing framework. The NAV discount directly addresses the appraisal-bias concern by marking the terminal position to secondary-market levels.

Limitations. The discounted NAV is still treated as a terminal cash flow and discounted back to the fund inception date, so the estimator retains sensitivity to the macro state at the dataset endpoint. The discount function d_i introduces model risk: it must be estimated or calibrated from secondary-market data that may not be representative of the broader PE universe, and it may vary with market conditions in ways that interact with the SDF specification.

6.2 Approach B: NAV-return SDF for all vintages

An alternative strategy abandons the cash-flow pricing restriction entirely and instead prices quarterly NAV returns for *all* vintages—seasoned and unseasoned alike. For each fund i with quarterly NAV observations, define $R^H_{t,i}$ as the gross quarterly Horizon-IRR return that solves

$$\mathrm{NAV}_{t-1,i}R + \sum_{\ell \in \mathcal{X}_{t,i}} X_{\ell,i} R^{1-\lambda_{\ell,t,i}} = \mathrm{NAV}_{t,i} + \sum_{\ell \in \mathcal{D}_{t,i}} D_{\ell,i} R^{1-\lambda_{\ell,t,i}},$$

where $\mathcal{X}_{t,i}$ and $\mathcal{D}_{t,i}$ are the dated capital calls and distributions inside the quarter, and $\lambda_{\ell,t,i} \in [0,1]$ is the fraction of the quarter elapsed when cash flow ℓ occurs. The one-period Euler pricing error is

$$e^H_{t,i}(\theta) = M_{t-1,t}(\theta) R^H_{t,i} - 1.$$

The fund-level moment is then a time-series average of these quarterly Euler errors, and the fold moment aggregates across funds and vintages as in Section 4. If only quarter-end NAVs and aggregate quarterly net cash flows are available, the simple return $(CF_{t,i} + \mathrm{NAV}_{t,i})/\mathrm{NAV}_{t-1,i}$ can be used as a coarse approximation, but the Horizon-IRR construction is the preferred definition because it accounts for intra-quarter cash-flow timing.

Advantages. The approach is internally consistent: the same pricing restriction applies to all vintages, eliminating the need for a seasoned/unseasoned partition or a NAV discount model. Each quarterly Euler error depends only on the one-period SDF, avoiding the exploding-alpha degeneracy entirely. This approach is closest in spirit to the NAV-return literature: Brown et al. [2023] nowcast fund NAVs, while Ghysels et al. [2025] construct high-frequency PE returns from NAV-based state-space models and use them for factor analysis and spanning tests.

Limitations. The approach relies on NAV quality for the *entire* sample, including seasoned vintages where realized cash flows are available and arguably preferable. For seasoned vintages, the cumulative CF-SDF restriction uses actual divestitures whose co-movement with public factors is genuine, whereas the NAV-return Euler restriction uses appraisals whose co-movement may be smoothed or biased. The approach therefore trades the unseasoned-vintage dilemma for a different concern: using appraisals where realized payoffs exist.

6.3 Approach C: hybrid cash-flow / normalized-Euler estimator (proposed)

The proposed approach uses the standard cumulative cash-flow pricing error for seasoned vintages and replaces it with a sequence of normalized one-period Euler restrictions on quarterly Horizon-IRR NAV returns for unseasoned vintages. Crucially, we do *not* discount those quarterly errors back to the initial date with the candidate cumulative SDF, because that would reconstruct the same long-horizon terminal-NAV exposure that Approach A retains. Instead, we average relative quarterly Euler errors within fund and then rescale the result by an exogenous duration factor chosen to make young-fund moments comparable in magnitude to seasoned NPV moments.

One-period Euler pricing error. For a fund i in an unseasoned vintage with quarterly NAV observations at dates $t = 0, 1, \dots, T_i$, compute the gross quarterly Horizon-IRR return $R^H_{t,i}$ from beginning NAV, ending NAV, and dated within-quarter cash flows as in Section 6.2. Define the relative one-period pricing error for quarter t as

$$e^H_{t,i}(\theta) = \Psi_{t-1,t}(\theta) R^H_{t,i} - 1,$$

where $\Psi_{t-1,t}(\theta)$ is the one-period stochastic discount factor. The conditional Euler restriction is

$$E\left[e^H_{t,i}(\theta_0) \mid \mathcal{F}_{t-1}\right] = 0.$$

This says that, conditional on information available at the start of the quarter, the Horizon-IRR return has unit risk-adjusted price under the true SDF. The conditional statement is important: predictable normalizations and instruments preserve the moment only under a conditional Euler restriction, not merely under an unconditional mean-zero restriction.

Normalization and scale matching. The Horizon-IRR Euler error is already a relative quarterly pricing error. Average it within fund:

$$\bar{r}_i(\theta) = \frac{1}{T_i} \sum_{t=1}^{T_i} e^H_{t,i}(\theta).$$

This object is a mean relative quarterly pricing error. To map $\bar{r}_i(\theta)$ into NPV-like units, let $s(i)$ denote the strategy of fund i, let $\bar{u}_s(a)$ be a benchmark age-a profile of NAV$/K$ for strategy s, and let q_{a-1} be a deterministic discount sequence (baseline $q_{a-1} = 1$, optional risk-free discounting as a robustness check). Define the duration-matching factor

$$\Lambda_i = c_{s(i)} \sum_{a=1}^{T_i} q_{a-1}\, \bar{u}_{s(i)}(a),$$

where $c_{s(i)}$ is an optional strategy-specific calibration constant. In the simplest implementation $c_{s(i)} = 1$. If the raw duration scale still leaves systematic size gaps, $c_{s(i)}$ can be chosen so that near-boundary seasoned and unseasoned funds have similar median absolute pricing moments within strategy. Because this last step touches realized pricing errors, $c_{s(i)}$ is then a nuisance parameter that must be calibrated on the estimation folds only and frozen for holdout scoring (the cross-fitting rule below); the baseline $c_{s(i)} = 1$ sidesteps the issue entirely by keeping Λ_i outside the SDF.

The normalized unseasoned fund moment is

$$g^U_i(\theta) = \Lambda_i\, \bar{r}_i(\theta),$$

and for an unseasoned vintage $v \in \mathcal{V}^U$ and fund cell $C_{n,v,r}$, define the cell-level pricing error as

$$\epsilon^U_{n,v,r}(\theta) = \frac{1}{|C_{n,v,r}|} \sum_{i \in C_{n,v,r}} g^U_i(\theta).$$

This normalization has a simple economic interpretation. The term $\bar{r}_i(\theta)$ is the average relative quarterly pricing error. The factor Λ_i is an externally chosen capital-at-risk duration: it measures how much NAV$/K$ exposure a typical fund of that age and strategy carries over its observed life. If the relative quarterly mispricing is roughly stable, $e^H_{t,i}(\theta) \approx \rho_i(\theta)$, then

$$g^U_i(\theta) \approx \rho_i(\theta)\, \Lambda_i,$$

so the young-fund moment is of the same order as a life-of-fund NPV error per committed dollar. This is the sense in which the normalization is "smart": it matches scales without using the candidate cumulative SDF itself.

Simple deterministic-duration robustness check. As a robustness check, one can avoid the fund-average-then-rescale structure and instead use a deterministic weighted sum of quarterly relative Euler errors:

$$g_i^{U,\mathrm{det}}(\theta) = \sum_{t=1}^{T_i} q_{t-1}\, \bar{u}_{s(i)}(t)\, e_{t,i}^{H}(\theta).$$

This is not the main specification, but it is useful because it removes the optional calibration constant c_s and makes the role of the duration profile transparent. If the preferred model changes only under this alternative, the empirical result is likely sensitive to scale matching rather than to the SDF's ability to price the underlying young-fund return moments.

Advantages. Realized cash flows are used where available (seasoned vintages); NAV-based moments are confined to unseasoned vintages where no alternative exists. The short-horizon Euler restriction avoids both the terminal-NAV discounting of Approach A and the full-sample NAV reliance of Approach B. The duration-matching normalization makes the young-fund block directly comparable to the seasoned NPV block.

Limitations. The approach introduces a seasoned/unseasoned boundary that is itself a design choice (the threshold δ), as well as the duration-matching normalizer Λ_i, which is calibrated outside the SDF. The resampling framework can test sensitivity to these choices by varying δ and the Λ_i specification and comparing holdout losses.

6.4 Letting the framework decide

The three approaches differ in how they treat unseasoned vintages, but all produce vintage-level pricing errors $\epsilon_{n,v,r}(\theta)$ that plug into the same fold-moment structure:

$$m_{n,r}(\theta; M, \eta) = \sum_{v \in A_{n,r}} \omega_{n,v,r}\, \epsilon_{n,v,r}(\theta; M, \eta).$$

The entire estimation and validation machinery of Section 5 applies without modification to each approach individually. Within a given approach, the resampling-based holdout loss provides a clear ranking of SDF models: the model with the lowest average holdout pricing loss is the one that best prices held-out moments *as defined by that approach*.

Cross-fitting nuisance choices. Any nuisance component used to build the unseasoned moments—the NAV discount d_i, the threshold δ if it is tuned, the benchmark duration profile $\bar{u}_s(a)$, the optional scale constant c_s, and any secondary-market NAV-discount model—should be estimated or selected using only the estimation side of the split. The fitted nuisance component is then frozen when computing holdout losses. In rolling-origin validation, the same rule is chronological: nuisance objects must use only information that would have been observable at the origin date τ.

Cross-approach comparability. Comparing holdout losses *across* approaches is less straightforward, because the three approaches define different moment conditions. Approach A evaluates a cumulative CF-SDF error with an adjusted terminal NAV; Approach B evaluates one-period Euler errors on NAV returns; Approach C evaluates a mixture of cumulative CF errors and normalized Euler errors. These moment conditions differ in scale, economic content, and sensitivity to the SDF parameters, so a holdout loss of, say, 0.02 under one approach and 0.05 under another does not imply that the first approach "works better"—the two numbers measure different things.

This is an important open question that requires careful resolution. One possible path is to separate *estimation* from *evaluation*: estimate $\hat{\theta}$ using each approach's own moments, but evaluate all three on a *common* criterion that does not depend on the approach. A natural candidate is the seasoned-vintage cash-flow pricing error, using the same terminal-NAV convention across approaches, since realized cash flows are the gold standard and all three approaches agree on their treatment of seasoned vintages. Each approach produces a different $\hat{\theta}_n^{(k)}$, but one can always evaluate

$$S_n^{\text{common}}(\hat{\theta}_n^{(k)}) = \frac{1}{|\mathcal{H}|}\sum_{r\in\mathcal{H}} L\left(\sum_{v\in\mathcal{V}^S\cap A_{n,r}} \omega_{n,v,r}\,\epsilon^S_{n,v,r}(\hat{\theta}_n^{(k)})\right), \qquad k\in\{A,B,C\},$$

asking: *which estimation approach produces the $\hat{\theta}$ that best prices held-out realized cash flows from seasoned vintages?* The rolling-origin design above is the sharper version of the same idea for unseasoned vintages: estimate at historical date τ, when the approaches genuinely differ in how they treat young funds, and evaluate later on realized cash-flow pricing errors after those funds have seasoned. We formalize this common criterion—and the principle that it must never contain a NAV term—in Section 6.5.

Diagnostics. To diagnose whether the estimator disproportionately fits one group, the empirical implementation should report separate holdout losses for $\mathcal{V}^S$ and $\mathcal{V}^U$:

$$S_n^S(M,\eta,\Pi) = \frac{1}{|\mathcal{H}|}\sum_{r\in\mathcal{H}} L\left(\sum_{v\in\mathcal{V}^S\cap A_{n,r}} \omega_{n,v,r}\,\epsilon^S_{n,v,r}(\hat{\theta}_n)\right),$$
$$S_n^U(M,\eta,\Pi) = \frac{1}{|\mathcal{H}|}\sum_{r\in\mathcal{H}} L\left(\sum_{v\in\mathcal{V}^U\cap A_{n,r}} \omega_{n,v,r}\,\epsilon^U_{n,v,r}(\hat{\theta}_n)\right).$$

If the average $\bar{S}^U$ is systematically lower than $\bar{S}^S$, the unseasoned-vintage restrictions are being trivially satisfied and contribute little to identification; if $\bar{S}^U \gg \bar{S}^S$, the short NAV histories are dominating the holdout loss and the model selection may be driven by NAV fit rather than cash-flow fit.

6.5 A NAV-free common evaluation target

The comparability problem has a clean resolution once we insist on one principle: *the metric used to compare approaches must never touch a NAV.* The three approaches differ precisely in how heavily they lean on appraisals, and appraisal quality is the very thing in doubt; any evaluation criterion that contains a NAV term therefore cannot adjudicate between them on neutral ground. Realized contributions and distributions, by contrast, are observed without appraisal error and are treated identically by all three approaches. They are the natural common currency, and the seasoned-only criterion S_n^{common} of the previous subsection is the special case that uses them.

The target. Fix an evaluation date τ' and let $\mathcal{R}_{\tau'}$ be the set of funds that are *realized* as of τ'—those whose residual NAV has fallen below the de minimis seasoning threshold δ. For such a fund the entire life is observed in realized cash flows, so its NAV-free pricing error is just the seasoned contribution g_i^S of Section 4.1 evaluated on a now-complete fund,

$$g_i^{\text{real}}(\theta) = \frac{1}{K_i}\sum_{t=0}^{T_i} \Psi_{0,t}(\theta)\,CF_{t,i}, \qquad i\in\mathcal{R}_{\tau'},$$

with any de minimis residual NAV handled by the *single* terminal-NAV convention shared across all approaches. No discounted appraisal enters. Aggregating to held-out cells exactly as in Section 5 gives, for each approach $k \in \{A, B, C\}$, the common score

$$S_n^{\text{real}}\left(\hat{\theta}_n^{(k)}\right) = \frac{1}{|\mathcal{H}|} \sum_{r \in \mathcal{H}} L\left(\sum_{v \in A_{n,r}} \omega_{n,v,r} \frac{1}{|C_{n,v,r}|} \sum_{i \in C_{n,v,r} \cap \mathcal{R}_{\tau'}} g_i^{\text{real}}\left(\hat{\theta}_n^{(k)}\right)\right).$$

Because the map $S_n^{\text{real}}(\cdot)$ is identical across approaches and free of NAVs, the three approaches differ only in how their parameters $\hat{\theta}_n^{(k)}$ were *estimated*, not in how they are *scored*. A value of 0.02 for one approach and 0.05 for another is now genuinely comparable: it asks which estimation route delivers the SDF that best prices fully realized cash flows. The principle is to state this as a design rule, not an option—evaluation is always NAV-free, regardless of which moments were used to estimate.

Two ways to make the held-out funds informative. The target discriminates between approaches only if the held-out realized funds were treated *differently* by the three approaches at estimation time; otherwise all approaches impose the same seasoned cash-flow restriction on those funds and the scores coincide by construction. Two designs achieve this.

- *Rolling origin (prospective).* This is the design of Section 5: estimate $\hat{\theta}_\tau^{(k)}$ using only information available at τ, when the target funds were still unseasoned and the approaches genuinely disagree about them, then score on g_i^{real} once those funds have realized at $\tau + \ell$. The rolling-origin score S_τ^{roll} is exactly S_n^{real} restricted to $\mathcal{V}_\tau^U \cap \mathcal{V}_{\tau+\ell}^S$.
- *Leave-future-out (retrospective).* Without rolling the information set back, hold out the *realized continuation* of currently-seasoned funds: estimate each approach using cash flows and NAVs only up to an elapsed fund age $\bar{a}$—so the approaches differ on the not-yet-realized tail—and score on the realized cash flows after $\bar{a}$. This uses the full panel and is cheaper than full rolling-origin re-estimation, at the cost of not being a real-time experiment.

In both designs the chronological cross-fitting rule of the previous subsection applies: every nuisance object must use only pre-evaluation information.

Backtesting the appraisal (auxiliary diagnostic). The same realized continuations support a direct, dollar-denominated test of the appraisal-bias concern that motivates the entire unseasoned-vintage problem. At a freeze date with elapsed quarters s, the model-implied value at s of the fund's later-realized cash flows is

$$\widehat{\text{NAV}}_{s,i}(\theta) = \frac{1}{\Psi_{0,s}(\theta)} \sum_{t>s} \Psi_{0,t}(\theta)\, CF_{t,i},$$

the cumulative-SDF price of the realized tail, renormalized to date s. The appraisal gap $\text{NAV}_{s,i} - \widehat{\text{NAV}}_{s,i}(\theta)$ has a benchmark term built only from realized flows. This is a diagnostic rather than the comparison metric—it asks whether an approach's fitted SDF rationalizes reported NAVs as fair risk-adjusted values—and it localizes *where* (which strategies, ages, or vintages) appraisals and model values diverge, which is exactly the smoothing/bias that Approaches A and B are most exposed to.

A selection caveat. S_n^{real} conditions on $i \in \mathcal{R}_{\tau'}$: only funds that have actually realized by the evaluation date enter the score, so slow-to-liquidate or lingering failed funds are under-represented and the target carries a mild survivorship/look-ahead selection. Two safeguards keep it honest. First, report the share of estimation-time funds that qualify for the target (coverage). Second, rely on the *paired* comparison of Section 11, in which every approach is scored on the *same* realized funds at the same origin τ; the selection then cancels in the cross-approach difference $S_n^{\text{real}}(\hat{\theta}_n^{(a)}) - S_n^{\text{real}}(\hat{\theta}_n^{(b)})$ even though it may bias the level of either score.

7 Identification and Stability

The exploding-alpha issue—where an arbitrarily large α drives cumulative discount factors to zero, creating a ridge of spurious near-minima—is a *finite-sample* pathology, not an asymptotic one. Under increasing-domain asymptotics indexed by vintage years, as $V \to \infty$ the cross-vintage moment averages over many vintage realizations of the cumulative SDF. These realizations are not independent—adjacent vintages overlap in calendar time and share macro and factor paths—so the relevant condition is weak dependence (mixing) across vintages rather than independence, and the limit should be read as a thin asymptotic (one new vintage per year, perhaps thirty to forty in total). Subject to this caveat, the global minimum of the population criterion $Q(\theta)$ is uniquely attained at θ_0 under standard identification conditions, and the spurious near-minima created by extreme alpha values vanish: no single alpha value can simultaneously zero out the discounted cash flows of many vintage realizations with heterogeneous timing and factor paths.

In finite samples, the exploding-alpha ridge must be handled explicitly. The useful safeguards are:

1. **Short-horizon Euler restrictions.** Under the hybrid approach C, unseasoned vintages contribute one-period Horizon-IRR Euler errors whose sensitivity to α is proportional to the quarter length Δt, not exponential in the fund horizon T. The duration-matching factor Λ_i is exogenous to the candidate cumulative SDF and does not reintroduce long-horizon alpha sensitivity. Under the full NAV-return approach B, all vintages use one-period discounting, eliminating long-horizon alpha sensitivity in the NAV-return block. Even under approach A, the NAV discount reduces the terminal position, partially mitigating the problem.
2. **Explicit alpha discipline.** The estimation problem should impose economically admissible parameter restrictions, such as positivity of the one-period SDF over the observed factor support and bounded alpha regions chosen before looking at the holdout score. The cumulative-alpha/direct-alpha parameterization in Section 8 is useful because it gives alpha an interpretable long-horizon abnormal-performance meaning, but it should still be accompanied by boundary-solution diagnostics.
3. **Resampling diagnostics, not resampling cure.** Validation folds do not mechanically penalize exploding alpha, because a degenerate cumulative SDF can also make held-out NPV moments small. Resampling is still useful because it reveals instability: across many sampled splits, weakly identified specifications should display high parameter dispersion, frequent boundary solutions, non-convergence, or sensitivity to small changes in moment construction.
4. **Asymptotic cash-flow dominance.** Under approaches A and C, only the most recent vintages within a bounded horizon D are classified as unseasoned. As the panel grows, $|\mathcal{V}^U|/V \to 0$, so the contribution of any NAV-based block to the cross-vintage average vanishes. The estimator is therefore asymptotically cash-flow dominated.

The initial-contribution anchor. The indexing convention interacts with the pathology. When the time-zero capital call enters the cumulative moment undiscounted ($\Psi_{0,0} = 1$), driving $\alpha \to \infty$ sends the seasoned NPV moment to $-X_{0,i} \neq 0$ rather than to zero, so the squared loss is bounded below by $X_{0,i}^2$ and the ridge is self-limiting. The degeneracy therefore reappears precisely when the undiscounted early cash flows are small relative to later ones—the typical PE shape—which is why we index seasoned cash flows from $t = 0$ throughout (Sections 4 and A) and treat the magnitude of the early-call anchor as a sample-specific diagnostic for exposure to the pathology.

Together, these mechanisms do not make exploding alpha disappear automatically. They make the pathology explicit, separate it from the model-selection role of validation, and give the empirical implementation concrete diagnostics for deciding when an SDF specification is too weakly identified to interpret.

NAV quality and the Euler moment condition. The conditional Euler moment $E[e^H_{t,i}(\theta_0) \mid \mathcal{F}_{t-1}] = 0$ makes a stronger claim: that the beginning-of-quarter information set prices the fund's Horizon-IRR return under the true SDF. For traded assets, no-arbitrage forces the analogous condition to hold. For private-equity funds, NAVs are appraisals whose co-movement with public factors may be smoothed or biased, so the Euler restriction holds only approximately. The paper's design controls this concern in three ways. First, under approaches A and C, only unseasoned vintages rely materially on NAV-based moments; seasoned vintages are priced with the chosen realized-cash-flow convention for any small residual NAV. Second, even within $\mathcal{V}^U$, the normalized Euler block uses one-quarter NAV return changes rather than a single large terminal NAV discounted back to inception. Third, the asymptotic irrelevance property ensures that the fraction of moment conditions relying on NAV quality vanishes as the panel grows. Under approach B, NAV quality affects the entire sample, which is the price paid for internal consistency.

8 Alpha Design

The resampling method does not by itself decide what alpha should mean. The cleanest baseline is to use a cumulative, NPV-style alpha rather than a period-by-period intercept alpha.

8.1 Baseline: cumulative alpha

For fold $F_{n,j}$, let $B_t(\beta)$ be the cumulative benchmark return implied by factor loadings β. A direct-alpha-style fold moment is

$$\psi_{n,j}(a,\beta) = \sum_t \frac{cf_{n,j,t}}{B_t(\beta)} \exp\bigl(-a\,\tau_{n,j,t}\bigr),$$

where $cf_{n,j,t}$ is the fold cash flow at time t, and $\tau_{n,j,t}$ is elapsed time from the fold origin to date t. For fixed β, the fold-implied cumulative alpha solves

$$\sum_t \frac{cf_{n,j,t}}{B_t(\beta)} \exp\bigl(-a_{n,j}(\beta)\,\tau_{n,j,t}\bigr) = 0.$$

This is close to the KN16/GPME/direct-alpha tradition because alpha is an NPV-style abnormal-performance object [Korteweg and Nagel, 2016, Gredil et al., 2023, Korteweg and Nagel, 2025]. It is also more natural for long-horizon PE cash flows than a monthly intercept alpha. The exponential $\exp(-a\,\tau)$ parameterizes alpha as a continuously compounded rate; for estimation this is equivalent to the discretely compounded $(1+\alpha)^{-t}$ up to the reparameterization $a = \log(1+\alpha)$.

The fold-implied alpha $a_{n,j}(\beta)$ serves a dual purpose in the resampling design. In *estimation*, a common alpha a is estimated jointly with β across the P estimation folds by minimizing the aggregate loss over fold moments $\psi_{n,j}(a, \beta)$. In *diagnostics*, the fold-specific implied alpha $a_{n,j}(\hat{\beta})$ is computed for each fold (including the validation fold) by solving $\psi_{n,j}(a, \hat{\beta}) = 0$ at the estimated $\hat{\beta}$. The cross-fold dispersion of $\{a_{n,j}\}_{j=1}^{P+h}$ is then a model diagnostic: a large spread indicates that the alpha estimate is sensitive to which funds enter the fold, and therefore that the aggregate alpha is weakly identified.

8.2 Alternative alpha definitions

The resampling framework can also compare alpha definitions as part of the design vector η: (i) the cumulative alpha baseline above, (ii) a DLP12-style intercept alpha, useful for comparison but vulnerable to exploding-alpha behavior in small or weakly identified samples [Driessen et al., 2012, 2008], and (iii) an error-term alpha ($\mathbb{E}[\epsilon]$) as in the machine-learning approach of Tausch and Pietz [2024]. The empirical question is not only which SDF factors perform best, but also which alpha definition yields stable and predictive PE pricing moments.

8.3 Identification-robust inference for α

Because exploding alpha is a weak-identification problem (Section 7), the natural inference tools are the weak-identification-robust statistics of the GMM literature rather than Wald-type confidence intervals, whose coverage is unreliable precisely when the alpha ridge is active. We recommend reporting a robust confidence set for α and, optionally, a single anchoring moment that restores its identification.

A profiled robust confidence set. Treat β (and any remaining parameters) as nuisance and profile them out at each hypothesized α. For a grid of values α_0, refit the remaining parameters on the estimation folds and evaluate the continuously-updated GMM objective $T_n(\alpha_0) = \min_\beta Q_n(\alpha_0, \beta)$. Under $H_0 : \alpha = \alpha_0$ an Anderson–Rubin / Stock–Wright S-statistic is asymptotically χ^2 regardless of identification strength [Stock and Wright, 2000], so inverting it,

$$\mathrm{CS}_{1-a}(\alpha) = \big\{\alpha_0 : \ T_n(\alpha_0) \leq c_{1-a}\big\},$$

yields a confidence set with correct coverage even on the ridge. The payoff is diagnostic as much as inferential: when α is weakly identified the set is wide or unbounded above, displaying the ridge honestly, whereas a Wald interval would report spurious precision. The profiled curve $T_n(\cdot)$ is also the cleanest picture of the exploding-alpha pathology, and it pairs with the fold-implied-alpha dispersion of Section 8: a flat right tail of T_n and a large spread of $\{a_{n,j}\}$ are two readings of the same weak identification.

A PME anchor that breaks the degeneracy. The NPV moment loses its grip on α in the large-α region because its sensitivity $\partial m^{\mathrm{NPV}}/\partial\alpha$ collapses as the cumulative discount factors vanish. The pooled-PME moment of Appendix A does not share this defect: being a ratio, it is invariant to a common rescaling of all discount factors and responds instead to the *relative* reweighting across maturities that α controls, so its α-sensitivity does not vanish on the ridge. Stacking a single pooled-PME moment (restricted to $\mathcal{V}^S$, where it is well defined) alongside the NPV moments therefore keeps the α-column of the moment Jacobian away from zero and locally re-identifies α, while the linear NPV moments continue to do the efficient work for β. Combined with the economically admissible restrictions of Section 7—one-period SDF positivity $1 + \alpha + \beta' f_t > 0$ over the observed factor support, which is linear in θ, and a pre-registered bounded-α box—this removes the inadmissible part of the ridge by construction and pins the admissible part with a moment that stays informative there.

9 Vintage-Stratified Factor Models

The resampling framework naturally accommodates time-varying factor exposures. Instead of estimating a single β across all vintages, define vintage-bucket-specific loadings β_b for vintage bucket $b \in \{B_1, \ldots, B_K\}$, where each bucket B_k collects vintages from a contiguous window (e.g., 2000–2004 buyout, 2015–2019 venture capital). The parameter vector becomes $\theta = (\alpha, \beta_{B_1}, \ldots, \beta_{B_K})'$, and the resampling framework applies without modification: more parameters require more estimation folds, and the holdout loss automatically penalizes overfitting.

Floating buckets. Overlapping vintage windows centered on each year—for example, a five-year window centered on 2010 that includes vintages 2008–2012—provide smooth estimates of how factor exposures evolve over time without sharp boundary effects. The bucket width is itself a tunable parameter that the resampling holdout loss can optimize.

Strategy-specific buckets. Factor exposures likely differ across PE strategies. A natural extension defines buckets jointly over vintage year and strategy (e.g., "2002 buyout factor," "2020 venture capital factor"), so that the holdout loss can compare: (i) pooled-β vs. bucket-β, (ii) narrow vs. wide bucket windows, and (iii) strategy-specific vs. pooled-strategy buckets.

Partial pooling and fused exposure paths. The free bucket model is easy to understand, but it can become noisy quickly: each additional vintage or strategy bucket adds parameters, and adjacent buckets may jump for sampling reasons rather than economic reasons. A more refined version keeps the bucket interpretation but regularizes the exposure path, in the spirit of regularized GMM for time-varying parameter models [Cui et al., 2024]. One option is hierarchical shrinkage,

$$\beta_b = \bar{\beta} + u_b, \qquad \sum_b \|u_b\|_2^2 \text{ penalized,}$$

which shrinks sparse or noisy buckets back toward a common exposure while allowing well-identified buckets to move away. Another option is a fused-lasso path penalty,

$$\lambda \sum_{b=2}^{K} \|\beta_b - \beta_{b-1}\|_1,$$

or its strategy-specific analogue, which encourages neighboring vintage buckets to share the same exposure unless the data strongly support a break. The advantage is a cleaner bias–variance tradeoff: the model can detect gradual time variation or discrete exposure regimes without estimating an unrelated β_b for every small cell. The tuning parameter λ becomes another design choice ranked by holdout loss.

This vintage-stratified specification showcases the flexibility of the resampling framework. The question of whether factor exposures are time-varying is empirically testable within the framework, not assumed a priori. A model with vintage-stratified betas that achieves lower holdout loss than a pooled-beta model provides direct evidence of time variation in PE factor exposures—a finding of independent interest for the asset-pricing literature.

Age-varying exposures along the fund life. Vintage buckets let factor loadings vary across cohorts, but PE risk arguably varies at least as much over a fund's *own* life. A young fund is illiquid, partly drawn, and carries conservatively marked positions—a low measured market exposure during the J-curve; as portfolio companies mature and approach exit, the position behaves more like levered public equity, with higher exposure. This motivates loadings that depend on fund age a rather than, or in addition to, vintage. Writing $a_{i,t}$ for the age of fund i at calendar quarter t, the one-period SDF becomes

$$M_{t-1,t}(\theta) = \frac{1}{1 + \alpha + \beta(a_{i,t})' f_t},$$

with $\beta(\cdot)$ a step function over age buckets or a smooth path regularized by the same fused-lasso or shrinkage penalty applied along the age axis. Identification comes from observing many funds at each age and from the within-fund time series of the Euler block, and the holdout loss arbitrates pooled-β vs. vintage-β vs. age-β vs. strategy-×-age-β just as before. A model with age-varying β that lowers the holdout loss is direct evidence of a life-cycle in PE factor exposure—an effect of independent economic interest and, plausibly, the first-order form of time variation.

An age–period–cohort caveat. Fund age, calendar period, and vintage are linearly dependent (period = vintage + age), the classic age–period–cohort collinearity, so one cannot freely include linear age, vintage, and calendar effects at once without a normalization. Here the macro "period" dimension is already carried by the factor realizations f_t inside the SDF, so age-β and vintage-β are separated only through the factor structure and the nonlinearity of the SDF, not through a free linear trend. The practical implication is to avoid stacking unrestricted linear vintage and age exposures simultaneously: identify one life-cycle axis at a time, or tie the two together with the shrinkage path above, and let the holdout loss decide whether the extra axis earns its parameters.

10 Literature Positioning

The paper can be positioned relative to the existing PE SDF literature along four dimensions:

- **From DLP12:** price private cash flows directly with an SDF [Driessen et al., 2012].
- **From KN16:** use aggregated pricing moments and an NPV-style abnormal-performance object [Korteweg and Nagel, 2016, 2025].
- **New here:** a general resampling-based model selection framework that treats the construction of PE pricing moments as a testable design choice. Instead of committing to one grouping, alpha definition, or unseasoned-vintage treatment, the framework varies these choices systematically and ranks them by holdout pricing loss.
- **Also new:** a three-competitor empirical horse race for unseasoned vintages—NAV-adjusted CF, full NAV returns, and hybrid normalized Euler—evaluated within a common resampling framework. The framework can further accommodate vintage-stratified factor models to test for time-varying risk exposures.

The paper is therefore not a repair of any existing SDF approach. It is a new empirical-design paper: how should we build, resample, and validate PE pricing moments when the sample is small and the moment construction is consequential? The resampling framework is the hero; individual design choices—moment type, alpha definition, unseasoned-vintage treatment, vintage-stratified betas—are its inputs.

11 Empirical Algorithm

For each candidate model M, alpha design η, unseasoned-vintage approach (A, B, or C), and partition scheme Π:

1. Partition vintages into seasoned ($\mathcal{V}^S$) and unseasoned ($\mathcal{V}^U$) sets (if using approach A or C).
2. Choose P estimation folds and h holdout folds.
3. Draw a sampled assignment n from Π.

4. Estimate or select nuisance objects using only the estimation folds (or only information available at the rolling-origin date).
5. Build fold cash-flow portfolios and fold pricing moments $m_{n,r}(\theta; M, \eta)$.
6. Estimate $\hat{\theta}_n(M, \eta, \Pi)$ on the P estimation folds.
7. Evaluate the fitted model on the holdout fold(s).
8. Repeat over many sampled assignments.
9. Rank model-design pairs by average holdout pricing loss.

Alongside the mean holdout loss, report diagnostics that help interpret the ranking:

- median and dispersion of holdout losses,
- dispersion of $\hat{\theta}_n$ across assignments,
- frequency of non-convergence,
- frequency of boundary or exploding-alpha solutions,
- separate holdout losses for $\mathcal{V}^S$ and $\mathcal{V}^U$,
- sensitivity of results to within-vintage resampling versus shuffled vintage-year portfolios.

11.1 Inference for score differences

Model rankings should be accompanied by uncertainty measures. For two model-design pairs a and b, compute their losses on the same sampled split n and form the paired difference

$$d_n(a, b) = S_n(a) - S_n(b).$$

The mean $\bar{d}(a, b)$ estimates the score gap, while the dispersion of $d_n(a, b)$ gives a standard error or bootstrap confidence interval for whether the gap is economically and statistically meaningful. Pairing matters because both models face the same split, vintage composition, factor realizations, and holdout funds; comparing paired differences removes a large amount of resampling noise. Since funds may be dependent within vintage years, strategies, or GP families, the bootstrap should resample at the relevant cluster level where identifiers are available (e.g., vintage, strategy-vintage cell, or GP). Boundary-alpha solutions should be reported as a paired diagnostic as well: if one model wins only because it frequently lands on the alpha boundary, that ranking is not economically interpretable.

The resampled-t statistic is anti-conservative. The naive standard error treats the N paired differences $\{d_n\}$ as independent and uses $\widehat{\text{Var}}(\bar{d}) = S_d^2/N$, where S_d^2 is their sample variance. They are not independent: every split reuses the same underlying funds, so the splits overlap heavily and $\text{Cov}(d_n, d_m) > 0$. As a consequence $S_d^2/N \to 0$ as the number of *draws* N grows with the fund sample held fixed, even though the genuine uncertainty—which lives in the finite, dependent cross-section of funds—does not shrink. Taken literally, the naive t-statistic can be pushed to any level of "significance" simply by drawing more partitions. This is the cross-validation variance problem of Bengio and Grandvalet [2004], who show that no unbiased estimator of the variance of a K-fold or resampled score exists.

A corrected resampled-t variance. A practical remedy is the corrected variance of Nadeau and Bengio [2003], which inflates the naive estimator by a term that does not vanish in N:

$$\widehat{\mathrm{Var}}_{\mathrm{corr}}(\bar{d}) = \left(\frac{1}{N} + \frac{n_{\mathrm{test}}}{n_{\mathrm{train}}}\right) S_d^2.$$

Here $n_{\mathrm{test}}/n_{\mathrm{train}}$ is the ratio of held-out to estimation observations, which in the balanced within-vintage design with P estimation and h holdout cells is simply h/P by fund count. The correction term $(h/P)\,S_d^2$ is the irreducible component the naive estimator omits: in the baseline $P = p_{\max}$, $h = 1$ it equals $S_d^2/p_{\max}$, so the corrected standard error converges to a positive limit rather than to zero. Paired model comparisons should be reported with this corrected t, not the naive one.

A nested cluster bootstrap. The corrected t still assumes exchangeable observations; PE funds are not, being dependent within vintage, strategy, and GP. The more faithful alternative is a nested cluster bootstrap that captures both the finite-sample fund uncertainty and this dependence: resample funds with replacement at the relevant cluster level (vintage, strategy-vintage cell, or GP), and for each bootstrap sample rerun the entire pipeline—redraw partitions, re-estimate $\hat{\theta}$, recompute the holdout scores—taking the distribution of $\bar{d}^*(a, b)$ across bootstrap replicates as the sampling distribution of the score gap. This is markedly more expensive, because it resamples over partitions *within* each bootstrap draw; the Nadeau and Bengio [2003] correction is the cheap analytic stand-in, and agreement between the two is itself reassuring. Finally, because the goal is to rank several models rather than to compare a single pair, the per-pair tests should be embedded in a multiplicity-aware procedure—reporting a model confidence set of statistically indistinguishable top models rather than a single "winner" whose lead is within the corrected standard error.

A Detailed Comparison of Pricing Moments

This appendix provides a detailed comparison of the dollar NPV and PME-ratio moment families. The main text uses the NPV moment as the baseline; the PME-ratio serves as a robustness check.

Dollar NPV moment. For seasoned funds in fold cell j in split n, aggregate discounted cash flows into one cumulative pricing error,

$$m_{n,j}^{\mathrm{NPV}}(\theta; M, \eta) = \sum_{i \in C_{n,j}} w_i \sum_{t=0}^{T_i} \Psi_{0,t}(\theta; M, \eta)\, CF_{t,i}, \qquad CF_{t,i} = D_{t,i} - X_{t,i},$$

where $X_{t,i}$ are contributions, $D_{t,i}$ are distributions, any small residual terminal NAV is handled according to the design convention in η, and $\Psi_{0,t}$ is the cumulative SDF. At the population parameter, $E[m_{n,j}^{\mathrm{NPV}}(\theta_0)] = 0$: the risk-adjusted NPV of fund cash flows vanishes.

Pros. (i) *Linearity* in cash flows yields tractable GMM derivatives, influence functions, and CLT-based inference. (ii) *Additivity* across funds makes the fold structure of Section 11, as well as subsampling and the bootstrap, transparent: the cell moment is the sum of fund-level moments, and the split moment is the sum of cell moments. (iii) *Compatibility with the normalized Euler NAV restriction* for unseasoned vintages: the seasoned block remains a linear cash-flow NPV, while the unseasoned block uses lagged-scale one-period Euler errors rescaled by an exogenous duration factor. Both blocks are expressed as per-committed-dollar pricing errors, and neither requires ratio denominators that can approach zero.

Cons. (i) *Scale-dependence.* A \$1B fund with a 1% relative NPV error contributes ten times more to the moment than a \$100M fund with the same relative error, so efficient GMM weighting implicitly tilts the estimator toward mega-funds. The influence function is mechanically heavy-tailed in the fund-size distribution rather than in genuine pricing disagreement. (ii) *Dollar-denominated comparisons across vintages* with heterogeneous aggregate commitments are not directly meaningful; normalization is left to the econometrician. (iii) *Sign cancellation.* A cross-sectional mean of zero is consistent with systematic under-pricing of one subgroup offsetting over-pricing of another, which the scalar moment cannot detect. This motivates the fine-grained fold structure but does not eliminate the concern within a fold.

PME-ratio moment. Define the risk-adjusted PME of fund i as

$$\mathrm{PME}_i(\theta; M, \eta) = \frac{\sum_t \Psi_{0,t}(\theta; M, \eta)\, D_{t,i}}{\sum_t \Psi_{0,t}(\theta; M, \eta)\, X_{t,i}},$$

from which two natural cell-level moments follow. The *average-PME* form is

$$m_{n,j}^{\mathrm{PME}}(\theta) = |C_{n,j}|^{-1} \sum_{i \in C_{n,j}} \big(\mathrm{PME}_i(\theta) - 1\big),$$

and the *pooled-PME* form is

$$m_{n,j}^{\text{PME-pool}}(\theta) = \frac{\sum_{i,t} \Psi_{0,t}(\theta) D_{t,i}}{\sum_{i,t} \Psi_{0,t}(\theta) X_{t,i}} - 1.$$

These two objects have different statistical meanings. Let

$$D_i^*(\theta) = \sum_t \Psi_{0,t}(\theta) D_{t,i}, \qquad X_i^*(\theta) = \sum_t \Psi_{0,t}(\theta) X_{t,i}, \qquad N_i^*(\theta) = D_i^*(\theta) - X_i^*(\theta).$$

The dollar-NPV restriction is $E[N_i^*(\theta_0)] = 0$. The average-PME population moment is

$$E\left[\frac{D_i^*(\theta_0)}{X_i^*(\theta_0)} - 1\right] = E\left[\frac{N_i^*(\theta_0)}{X_i^*(\theta_0)}\right] = \mathrm{Cov}\left(N_i^*(\theta_0), \frac{1}{X_i^*(\theta_0)}\right),$$

where the last equality uses $E[N_i^*(\theta_0)] = 0$. Thus NPV pricing does not generally imply that the average PME has mean one; it does so only under stronger conditions, for example $E[N_i^*(\theta_0) \mid X_i^*(\theta_0)] = 0$. By contrast, the population pooled-PME object is

$$\frac{E[D_i^*(\theta_0)]}{E[X_i^*(\theta_0)]} - 1 = \frac{E[N_i^*(\theta_0)]}{E[X_i^*(\theta_0)]},$$

so it is aligned with the dollar-NPV restriction at the population level provided $E[X_i^*(\theta_0)] > 0$. The sample pooled PME is still a random ratio and can be biased in small samples, but it converges to a ratio of means rather than to the mean of individual ratios.

Pros. (i) *Scale-invariance.* The ratio is unit-free, so fund-size heterogeneity does not mechanically up-weight mega-funds. A \$100M fund and a \$1B fund enter the cross-sectional moment on equal footing. (ii) *Practitioner interpretability.* PME is the native performance metric of the industry; reporting results in the same units aids dissemination. (iii) *Bounded downside.* The fund-level PME is bounded below by zero, so $\mathrm{PME}_i - 1$ is bounded below by -1: a single disaster fund cannot dominate the moment the way it can dominate a dollar NPV.

Cons. (i) *Nonlinearity.* The moment is a ratio in θ; its derivative involves the ratio of two correlated sums and is numerically fragile when the denominator (discounted contributions) is small. This occurs mechanically for young funds with back-loaded capital calls — precisely the vintages that the Euler NAV restriction is designed to retain. (ii) *Mean-of-ratios versus ratio-of-means.* The average-PME and pooled-PME moments need not agree even asymptotically. Average PME estimates $E[D_i^*/X_i^*]$, while pooled PME estimates $E[D_i^*]/E[X_i^*]$. The gap is an economic weighting choice, not a minor finite-sample nuisance: average PME gives each fund equal weight, whereas pooled PME weights funds by discounted contributions. (iii) *No clean short-horizon analogue for unseasoned vintages.* Integrating unseasoned vintages $\mathcal{V}^U$ into a PME framework typically requires treating the latest $\text{NAV}_{T_i,i}$ as a terminal distribution and discounting it back to the fund inception date — exactly the long-horizon discounting mechanism that the normalized Euler block is designed to avoid. (iv) *Inference.* Upper-tail skewness in PME_i (funds can return many times their capital, but cannot lose more than it) distorts CLT-based standard errors and t-statistics unless explicitly robustified, e.g., by bootstrapping the cell-level moment or by employing median-based loss functions.

B Why We Do Not Discount Euler Errors to Time Zero

Let

$$\eta_{t,i}^{\$}(\theta) = \Psi_{t-1,t}(\theta)\big(CF_{t,i} + \text{NAV}_{t,i}\big) - \text{NAV}_{t-1,i}$$

denote the dollar NAV Euler error corresponding to the quarter-end net-cash-flow approximation. Multiplying $\eta_{t,i}^{\$}(\theta)$ by $\Psi_{0,t-1}(\theta)$ and summing would yield a valid moment, and for fully liquidated funds it would telescope. But for the young funds of actual interest it would recreate the long-horizon terminal-NAV term:

$$\sum_{t=1}^{T_i} \Psi_{0,t-1}(\theta)\,\eta_{t,i}^{\$}(\theta) = \sum_{t=1}^{T_i} \Psi_{0,t}(\theta)\,CF_{t,i} \;+\; \Psi_{0,T_i}(\theta)\,\text{NAV}_{T_i,i} \;-\; \text{NAV}_{0,i}.$$

For an unseasoned fund, $\text{NAV}_{T_i,i} \neq 0$, so discounting the quarterly Euler errors back to time zero reconstructs exactly the large terminal-NAV present-value term that generates instability. The normalized construction in Section 6.3 sacrifices path-by-path telescoping in order to preserve the short-horizon dependence of the young-fund block on the candidate SDF.

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